

Comparison of Lightweight Encryption Algorithms

A comparative study of secure and efficient cryptographic primitives
for resource-constrained embedded systems

Final Project Report Submitted

In partial fulfillment of the requirements for the award of the
Bachelor of Technology

Authors

Rajesh Kumar Jena (B122088)

Soumil Kumar (B122114)

Punit Kumar (B422041)

Under the supervision of

Prof. Debasish Jena



International Institute of Information Technology, Bhubaneswar

May 2026

Approval of the Viva Voce Board

May 19, 2026

Certified that the report entitled “Comparison of Lightweight Encryption Algorithms,” submitted by Rajesh Kumar Jena (B122088), Soumil Kumar (B122114), and Punit Kumar (B422041) to the International Institute of Information Technology, Bhubaneswar, in partial fulfillment of the B.Tech programme requirements for the award of the Bachelor of Technology, has been accepted by the examiners during the viva voce examination held today.

Prof. Debasish Jena (Supervisor)

External Examiner

Certificate

This is to certify that the report entitled “Comparison of Lightweight Encryption Algorithms,” submitted by Rajesh Kumar Jena (B122088), Soumil Kumar (B122114), and Punit Kumar (B422041) to the International Institute of Information Technology, Bhubaneswar, is a record of bona fide work carried out under my supervision. The report is submitted for the end-semester evaluation of the B.Tech programme.

Prof. Debasish Jena

Declaration

We hereby declare that:

1. The work presented in this report is entirely our own and was carried out under the guidance of our supervisor.
2. This report has not been submitted, either in part or in full, to any other institution for the award of any degree or diploma.
3. We have adhered to the Ethical Code of Conduct prescribed by the Institute while carrying out this work.
4. Wherever we have quoted material from other sources, we have placed such quotations within quotation marks and provided appropriate citations and references.
5. Whenever we have used ideas, data, analysis, or text derived from other sources, we have duly acknowledged them within the report and included full details in the reference section.
6. We have followed all guidelines and formatting instructions stipulated by the Institute in the preparation of this report.

Rajesh Kumar Jena

B122088

Soumil Kumar

B122114

Punit Kumar

B422041

Acknowledgments

We express our sincere gratitude to our supervisor, Prof. Debasish Jena, for his invaluable guidance, continuous support, and insightful feedback throughout this project. His expertise was instrumental in helping us navigate challenges and refine our work.

We also thank the faculty and staff of the Department of Computer Science and Engineering and the Department of Information Technology at the International Institute of Information Technology, Bhubaneswar, for providing the resources, infrastructure, and academic environment necessary for the successful completion of this project.

Finally, we extend our heartfelt appreciation to our families and friends for their constant support and understanding throughout the duration of this work.

Rajesh Kumar Jena

Soumil Kumar

Punit Kumar

Abstract

The proliferation of resource-constrained devices in IoT networks, embedded controllers, and sensor systems has increased the need for lightweight cryptographic solutions. Conventional encryption algorithms often exceed the computational capabilities, memory constraints, and energy budgets of such platforms. This study presents a comparative analysis of three prominent lightweight encryption algorithms—PRESENT, SPECK, and ASCON—and evaluates their suitability for constrained embedded environments.

The methodology combines theoretical analysis of design principles with empirical performance benchmarking on a Raspberry Pi Pico microcontroller using Rust implementations. Metrics, including execution time, memory utilization, and throughput, are systematically evaluated.

The findings provide a clearer understanding of how lightweight ciphers balance security and efficiency in practical embedded systems, revealing clear performance hierarchies with significant variations in throughput, memory footprint, and security properties that inform practical deployment decisions.

Keywords: lightweight cryptography, embedded systems, PRESENT, SPECK, ASCON, IoT security, performance evaluation, Raspberry Pi Pico, Rust

Contents

Approval of the Viva Voce Board	i
Certificate	ii
Declaration	iii
Acknowledgments	iv
Abstract	v
List of Abbreviations	ix
Introduction	1
1 Introduction	1
1.1 Historical Context	1
1.2 Motivation: Computing at the Edge	1
1.3 Problem Statement	2
1.4 Objectives	2
2 Literature Survey	4
2.1 Evolution of Lightweight Cryptography	4
2.2 Prior Benchmarking Studies	5
3 Experimental Methodology	7
3.1 Algorithms Evaluated	7
3.2 Performance Metrics	9
3.3 Implementation Plan	9

4	Results and Discussion	12
4.1	Experimental Setup	12
4.2	Experimental Results	12
4.2.1	Block Cipher Performance	12
4.2.2	Mode Comparison	13
4.2.3	Energy Analysis	14
4.2.4	ASCON Phase Breakdown	16
4.3	Memory Footprint Analysis	17
4.4	Speedup Analysis	17
4.5	Discussion	19
5	Conclusion	22
A	Test Vectors	25

List of Tables

2.1	Comparison of Prior Benchmarking Studies	5
3.1	Design Properties Comparison	7
3.2	SPECK family parameter set for all block and key size variants	8
4.1	Block Cipher Benchmarks on Raspberry Pi Pico	13
4.2	Energy Consumption (μ J) by Payload Size on Raspberry Pi Pico (estimated at 80–90 mA active current)	15
4.3	Mode Comparison (64-byte blocks) on Raspberry Pi Pico	16
4.4	ASCON AEAD Phase Breakdown	16
4.5	Memory Footprint Comparison	17
4.6	Speedup vs ASCON at Different Payload Sizes	18
5.1	Scorecard Comparison: PRESENT, SPECK, ASCON	22

A.1	Test vectors for all implemented algorithms	25
-----	---	----

List of Figures

4.1	Block Cipher Throughput Comparison (RP2040 Benchmark)	13
4.2	Mode Comparison: ECB vs CBC vs CTR Performance	13
4.3	Power Consumption Analysis: Energy vs Payload Size (RP2040)	15
4.4	Scaling Analysis: Performance vs Payload Size (RP2040)	18
4.5	Performance Comparison: PRESENT, SPECK, ASCON on Raspberry Pi Pico	18

List of Abbreviations

Abbreviation	Full Form
AEAD	Authenticated Encryption with Associated Data
ARM	Advanced RISC Machines
ARX	Add-Rotate-XOR
ASCII	American Standard Code for Information Interchange
CBC	Cipher Block Chaining
CPB	Cycles Per Byte
CTR	Counter Mode
ECB	Electronic Codebook
FIPS	Federal Information Processing Standards
IoT	Internet of Things
ISO	International Organization for Standardization
NIST	National Institute of Standards and Technology
PRESENT	PRESENT Block Cipher
RAM	Random Access Memory
RFID	Radio Frequency Identification
RISC	Reduced Instruction Set Computer
ROM	Read-Only Memory
SPECK	Simon and Speck Block Cipher
SPN	Substitution-Permutation Network

1. Introduction

1.1 Historical Context

Cryptography has evolved through several distinct eras, each driven by the constraints of its time. Early manual ciphers—such as the Caesar cipher, the Vigenère cipher, and the Enigma machine—relied on simple substitution and transposition operations feasible by hand or with mechanical assistance [1]. The advent of digital computing in the mid-20th century enabled far stronger encryption, leading to standardized algorithms such as the Data Encryption Standard (DES) in 1977 and the Advanced Encryption Standard (AES) in 2001 [2]. These algorithms were designed for desktop and server-class hardware with ample computational resources.

The proliferation of the Internet of Things (IoT) introduced a new class of devices—microcontrollers with limited processing power, memory, and energy budgets—for which traditional algorithms proved impractical. AES, while extremely secure, requires significant memory for its S-box tables and performs poorly on 8-bit and low-end 32-bit processors without hardware acceleration. This gap motivated the development of *lightweight cryptography*: algorithms engineered specifically for constrained environments while maintaining practical security levels. Standards bodies responded accordingly: ISO/IEC standardized PRESENT as ISO/IEC 29192-2 in 2019, and NIST selected ASCON as the Lightweight Cryptography Standard (SP 800-232) in 2023 [10, 6], marking the formal recognition of lightweight primitives in the cryptographic landscape.

1.2 Motivation: Computing at the Edge

The rapid expansion of Internet of Things (IoT) deployments has resulted in billions of devices operating under strict constraints on processing capability, memory capacity, and energy resources. Traditional cryptographic algorithms, designed for powerful general-purpose systems, frequently prove impractical in such environments.

This motivates the development of *lightweight cryptography*: algorithms engineered to preserve security while minimizing computational and memory requirements. Lightweight designs emphasize:

- minimization of computational complexity and cycle counts,
- reduction of RAM and ROM footprint,
- efficient operation on low-power hardware, and
- maintenance of practical security levels (typically 80–128 bits).

Accordingly, this investigation explores lightweight encryption algorithms as effective solutions for securing communication on embedded devices such as the Raspberry Pi Pico.

1.3 Problem Statement

Conventional encryption and public-key algorithms such as AES, RSA, and ECC offer strong security guarantees but were developed with the assumption of abundant computational and memory resources. Their requirements often exceed the capabilities of microcontrollers commonly used in IoT devices [8].

Devices with limited processing frequencies, small RAM capacities, and stringent power budgets struggle to run cryptographic routines that depend on large lookup tables, multi-precision arithmetic, or high cycle counts [8]. As a result, developers sometimes deploy weakened, partial, or ad hoc security mechanisms, leading to vulnerabilities such as unencrypted communication, reused keys, and insufficient authentication. Incidents such as the Mirai botnet attack highlight the risks of insecure embedded systems [11].

Lightweight cryptography addresses this gap by offering efficient primitives tailored to constrained platforms without compromising core security goals.

1.4 Objectives

This report aims to:

- examine the design principles of lightweight cryptographic algorithms,
- evaluate selected lightweight ciphers (PRESENT, SPECK, ASCON) in terms of execution time, memory usage, and implementation cost,

- implement and benchmark these algorithms on a Raspberry Pi Pico using Rust [12], and
- identify design trade-offs and assess their suitability for real-world constrained environments.

Several prior studies have benchmarked lightweight ciphers on embedded platforms, including comparative evaluations of PRESENT, SPECK, and SIMON on ARM Cortex-M series microcontrollers [4, 7]. However, to our knowledge, no prior work provides a side-by-side comparison of PRESENT, SPECK, and ASCON on the RP2040 platform using Rust implementations. This study fills that gap, contributing: (i) a reproducible Rust-based benchmarking framework for the Raspberry Pi Pico, (ii) empirical performance data across block cipher, mode, and payload-size dimensions, and (iii) practical deployment recommendations based on measured trade-offs.

2. Literature Survey

2.1 Evolution of Lightweight Cryptography

The need for lightweight cryptographic primitives was first articulated in the early 2000s as RFID tags and sensor networks began to proliferate. Traditional block ciphers such as AES, while offering strong security, require 2,500–3,000 gate equivalents (GE) for a hardware implementation, exceeding the $\sim 2,000$ GE budget typical of passive RFID tags [7]. This motivated the development of dedicated lightweight designs.

The first generation of lightweight ciphers emerged between 2007 and 2013, led by PRESENT [3] and the NSA-designed SIMON/SPECK family [4]. These focused primarily on block cipher primitives with reduced gate counts and simplified round functions. PRESENT achieved approximately 1,570 GE—nearly half the cost of AES—by using a 4-bit S-box and a bit-permutation layer implementable as simple wire crossings in hardware. SPECK targeted a complementary niche: software efficiency on general-purpose microcontrollers, using only ARX operations that map efficiently to most instruction sets.

The second generation, marked by ASCON’s submission to the CAESAR competition in 2014 [5], shifted focus toward authenticated encryption (AEAD) rather than standalone encryption. This reflected the practical reality that most IoT deployments require both confidentiality and integrity. ASCON’s sponge-duplex construction provides both in a single primitive, unlike PRESENT and SPECK which require a separate MAC or hash for authentication. In 2023, NIST formalized this evolution by selecting ASCON as the Lightweight Cryptography Standard (SP 800-232) [6], cementing its role as the recommended AEAD primitive for constrained devices.

The NIST lightweight cryptography standardization process, initiated in 2015 with the publication of NISTIR 8114 [7], evaluated 57 candidate algorithms across multiple dimensions including security, performance, and implementation cost. After a multi-round selection process involving hardware and software benchmarking, ASCON was announced as the winner in February 2023 and subsequently published as NIST SP 800-

232 in 2025. This process established a rigorous framework for evaluating lightweight primitives and provided the cryptographic community with a standardized benchmark for future designs.

2.2 Prior Benchmarking Studies

Several studies have benchmarked lightweight ciphers on embedded platforms prior to this work. Table 2.1 summarises the most relevant studies and the gap each leaves, which this work addresses.

Table 2.1: Comparison of Prior Benchmarking Studies

Study	Year	Platform	Algorithms	Metrics & Gap Left
Beaulieu et al. [4]	2015	8-bit AVR, Cortex-M3	SPECK family	CPB only; older MCUs, no ASCON/PRESENT comparison
NISTIR 8114 [7]	2017	FPGA, ASIC (multiple)	57 candidates	Throughput, GE; hardware focus, no Cortex-M0+ software data
Honnnavalli et al. [13]	2024	IoT boards (Cortex-M)	AES, SPECK, ASCON	Throughput, Energy; no RP2040, no PRESENT, no mode analysis
Kaur et al. [14]	2025	Various	ASCON only	Security/attack survey; no empirical performance benchmarking
This Work	2026	RP2040 (Cortex-M0+)	PRESENT, SPECK, ASCON	Throughput, Memory, Energy, Latency across modes; first Rust-based RP2040 comparison of all three primitives

Beaulieu et al. reported SPECK-64/128 at 164 CPB on 8-bit AVR and approximately 16 CPB on 32-bit Cortex-M3 [4], demonstrating the cipher’s suitability across architectures but leaving the Cortex-M0+ family uncharacterised. The NISTIR 8114 survey [7] evaluated 57 candidates and identified PRESENT as the most hardware-efficient option, but noted its poor software performance and highlighted the need for benchmarking on contemporary microcontrollers. Honnavalli et al. (2024) compared AES-128, SPECK, and ASCON on IoT boards, finding SPECK superior in throughput and ASCON more energy-efficient [13], but did not include PRESENT or the RP2040 platform. Kaur et al. (2025) provide the first comprehensive survey of the NIST ASCON standard covering attacks and countermeasures [14], but offer no empirical performance data.

Other notable studies include the ISO/IEC standardisation of PRESENT (ISO/IEC 29192-2:2019) [10] and the ongoing cryptanalysis of the SPECK family [9], which have established confidence in the security margins of these algorithms. No prior study provides a

side-by-side comparison of PRESENT, SPECK, and ASCON on the RP2040 using Rust implementations—the gap this work addresses.

3. Experimental Methodology

3.1 Algorithms Evaluated

We selected three lightweight cryptographic algorithms for evaluation: PRESENT, SPECK, and ASCON.

Property	PRESENT	SPECK	ASCON
Block Size	64 bits	64 bits	64 bits
Key Sizes	80, 128 bits	64, 96, 128, 144, 192, 256 bits	128 bits
Structure	SPN	ARX Feistel	Sponge-Duplex
Rounds	31/32	22-34	12
Year	2007	2013	2014
Standardization	ISO/IEC 29192-2	—	NIST SP 800-232
AEAD Support	No	No	Yes

Table 3.1: Design Properties Comparison

PRESENT is a 64-bit Substitution-Permutation Network (SPN) block cipher with 80/128-bit key options, designed for ultra-lightweight hardware implementations [3]. Each round applies AddRoundKey, a substitution layer of sixteen parallel 4-bit S-boxes, and a bit-wise permutation for diffusion. Its gate count of approximately 1,570 GE makes it one of the most compact ciphers standardized in ISO/IEC 29192-2, suitable for RFID and sensor nodes. Its 4-bit S-box minimizes differential and linear propagation, with a maximum differential probability of 2^{-2} and maximum linear bias of 2^{-2} , providing provable security bounds over the full 31 rounds.

SPECK is a family of Add-Rotate-XOR (ARX)-based block ciphers optimized for software efficiency on microcontrollers [4]. Its Feistel-like structure uses only add, rotate, and XOR operations on two n -bit words per round, with rotation constants α and β varying by variant. The SPECK family encompasses multiple block and key sizes, each with specific rotation parameters and round counts:

Variant	Block size	Key size	Rounds	α, β
SPECK-32/64	32	64	22	7, 2
SPECK-48/72	48	72	22	8, 3
SPECK-48/96	48	96	23	8, 3
SPECK-64/96	64	96	26	8, 3
SPECK-64/128	64	128	27	8, 3
SPECK-96/144	96	144	28	8, 3
SPECK-96/192	96	192	29	8, 3
SPECK-128/192	128	192	30	8, 3
SPECK-128/256	128	256	34	8, 3

Table 3.2: SPECK family parameter set for all block and key size variants

This design makes it one of the fastest lightweight ciphers in software. Despite scrutiny regarding its NSA origin and ISO rejection [9], SPECK remains widely studied and deployed due to its exceptional software performance. Unlike SPN ciphers, SPECK has no provable differential or linear bounds; its security relies on the interaction of modular addition, rotation, and XOR. Best published attacks reduce up to 20 of 27 rounds for SPECK-64/128, leaving a practical security margin in the full-round cipher.

ASCON is a sponge-duplex Authenticated Encryption with Associated Data (AEAD) scheme with a 320-bit internal state (five 64-bit words), selected as the National Institute of Standards and Technology (NIST) Lightweight Cryptography Standard (SP 800-232) in 2023 [5, 6]. The ASCON family includes three variants: **ASCON-128** (primary variant with 64-bit rate, 256-bit capacity, 128-bit security), **ASCON-128a** (optimized for higher throughput with 128-bit rate, 192-bit capacity, 128-bit security, and 8 intermediate rounds; not evaluated in this study), and **ASCON-80pq** (designed for quantum resistance with 160-bit key; not evaluated in this study). This work focuses exclusively on ASCON-128. The ASCON-128 variant uses two permutation depths: p^a (12 rounds) for initialization and finalization, and p^b (6 rounds) for intermediate processing. Each permutation round operates on the 320-bit state $S = S_0 || S_1 || S_2 || S_3 || S_4$ through three steps:

1. **Add constants:** the round constant c_r is XORed into S_2 ,
2. **Substitution layer:** 64 parallel 5-bit S-boxes are applied across the five 64-bit words using a diffusion-optimized mapping,
3. **Diffusion layer:** a linear transformation based on Σ -type shifts (right-rotate by 19 and 28 for word S_0 , by 61 and 39 for S_1 , by 1 and 6 for S_2 , by 10 and 17 for

S_3 , and by 7 and 41 for S_4) ensures full-state diffusion within a small number of rounds.

The 64-bit rate (ASCON-128) and 256-bit capacity split provides 128-bit security while enabling built-in authenticated encryption and hashing in a single primitive, suitable for security-critical IoT deployments. (ASCON-128a provides higher throughput at the cost of reduced capacity.) The 256-bit capacity provides 128-bit security with proven indistinguishability bounds for the sponge-duplex construction, and extensive cryptanalysis has found no significant attacks on the full 12-round permutation.

3.2 Performance Metrics

Performance evaluation focuses on execution efficiency on the ARM Cortex-M0+ micro-controller:

- **Execution Time:** measured using the RP2040’s hardware timer (`time_us_64()`) with interrupts disabled; cycles derived as *time in μs* $\times 133$.
- **Cycles Per Byte (CPB):** $CPB = \frac{total\ cycles}{total\ bytes\ processed}$.
- **Memory Utilization:** ROM footprint via `cargo size -release`, SRAM via ELF `.bss/.data` sections.
- **Throughput:** measured by encrypting fixed-size buffers.

3.3 Implementation Plan

All experiments used Rust (`no_std`) on a Raspberry Pi Pico (ARM Cortex-M0+, 133 MHz, 264 KB SRAM, 2 MB Flash), with `rp2040-hal` for hardware timer access and `probe-rs/cargo-embed` for debugging. Build configuration: release mode with LTO and size optimization.

Rust Implementation Details

Each cipher was implemented as a standalone Rust crate targeting the `thumbv6m-none-eabi` target triple (ARM Cortex-M0+). Each cipher provides `encrypt_block()` and `decrypt_block()` methods (and mode implementations like ECB, CBC, CTR where applicable) on its cipher struct, enabling consistent benchmarking across algorithms.

For PRESENT, the 4-bit S-box was implemented as a 16-entry lookup table (`const` array in ROM), and the bit-permutation layer was implemented using bitwise shift-and-mask operations across the two 32-bit halves of the 64-bit state. The key schedule for PRESENT-80 and PRESENT-128 was computed on-the-fly during encryption to minimize RAM usage.

SPECK’s ARX round function was implemented using Rust’s native `rotate_left()`, `rotate_right()`, `wrapping_add()`, and `xor` operators, which the LLVM backend compiles directly to single-cycle ARM `RORS`, `ROLS`, `ADDS`, and `EORS` instructions. The round keys were precomputed and stored in a fixed-size array on the stack. For SPECK-64 variants, the implementation supports 96-bit keys (SPECK-64/96) and 128-bit keys (SPECK-64/128), with the key schedule automatically adapting the round count (26 rounds for 96-bit keys, 27 rounds for 128-bit keys).

ASCON’s 320-bit state was represented as a `[u64; 5]` array. The substitution layer uses a 5-bit S-box implemented via bit-slicing techniques, and the diffusion layer applies cyclic rotations on each 64-bit word. The permutation variants differ by variant: ASCON-128 and ASCON-80pq use p^a (12 rounds) for initialization/finalization and p^b (6 rounds) for processing, while ASCON-128a uses p^a (12 rounds) for initialization/finalization and an 8-round permutation for intermediate processing to enable higher throughput at reduced capacity.

Energy Measurement Methodology

Energy consumption was estimated from cycle-count measurements using the standard physical energy model: $E = V \times I \times t$, where time is derived from the cycle counter (dividing cycles by the CPU frequency). The RP2040 operating voltage was fixed at 3.3 V (standard operating voltage), and active current was set to 80–90 mA based on RP2040 datasheet active-mode specifications during high-frequency cryptographic workloads at full CPU utilization (133 MHz). This gives an estimated energy per cycle of approximately 1.92–2.16 nJ. **Important caveat:** Energy values reported herein represent relative estimates derived from cycle counts and should be interpreted primarily for *comparative purposes* (e.g., cipher-to-cipher ratios) rather than absolute power consumption. Precise energy profiling would require direct instrumentation with a current probe or power analyzer. Energy estimates were calculated for each benchmark run using the cycle-count data reported in throughput analysis, ensuring consistency between timing and energy metrics. Initialization overhead was captured separately by measuring the

fixed per-operation costs (independent of payload size) for each cipher mode, allowing separation of algorithmic efficiency from platform overheads.

Benchmarking procedure: hardware timer for cycle-accurate timing; minimum 1000 iterations with warm-up runs; single-block and multi-block measurements; ASCON phases (init, absorb, encrypt, finalize) evaluated separately. Memory profiling used `cargo bloat` and `cargo size -release - -A` for code size and section analysis.

We verified implementations against official test vectors from algorithm specifications and cross-validated against known reference implementations.

4. Results and Discussion

This chapter presents the experimental results from benchmarking the lightweight cryptographic algorithms on the Raspberry Pi Pico. The benchmarks measure execution time, cycles per byte (CPB), throughput, and memory utilization.

4.1 Experimental Setup

All benchmarks followed the procedures described in Chapter 2 (hardware timer, 1000+ iterations, interrupts disabled, warm-up runs, cross-validation with test vectors). Variation across runs was less than 2%; reported values are means.

4.2 Experimental Results

All benchmarks were conducted on the Raspberry Pi Pico (ARM Cortex-M0+, 133 MHz, 264 KB SRAM) as the target embedded platform. All measurements used the hardware timer with interrupts disabled for accurate cycle counts.

4.2.1 Block Cipher Performance

Table 4.1 presents single-block encryption performance on the Raspberry Pi Pico. SPECK achieved the highest throughput (23.67 Mbps on Pico), approximately 70x faster than PRESENT (0.34 Mbps) and 3x faster than ASCON (7.56 Mbps). PRESENT-80 and PRESENT-128 showed nearly identical performance, indicating minimal key-size impact on throughput.

Algorithm	Raspberry Pi Pico (ARM Cortex-M0+)		
	ns/block	CPB	Mbps
PRESENT-80	23,665	2,958	0.34
PRESENT-128	24,542	3,068	0.33
SPECK-64/96	336	42	23.81
SPECK-64/128	338	42	23.67
ASCON-128	2,117	132	7.56

Table 4.1: Block Cipher Benchmarks on Raspberry Pi Pico

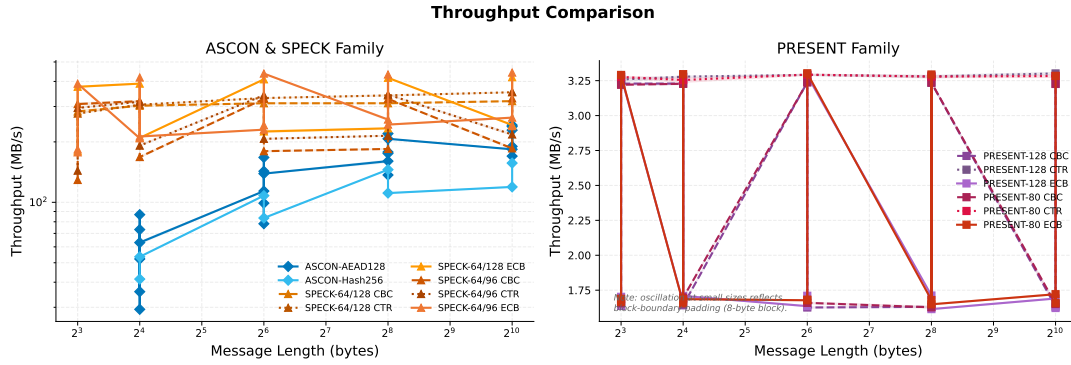


Figure 4.1: Block Cipher Throughput Comparison (RP2040 Benchmark)

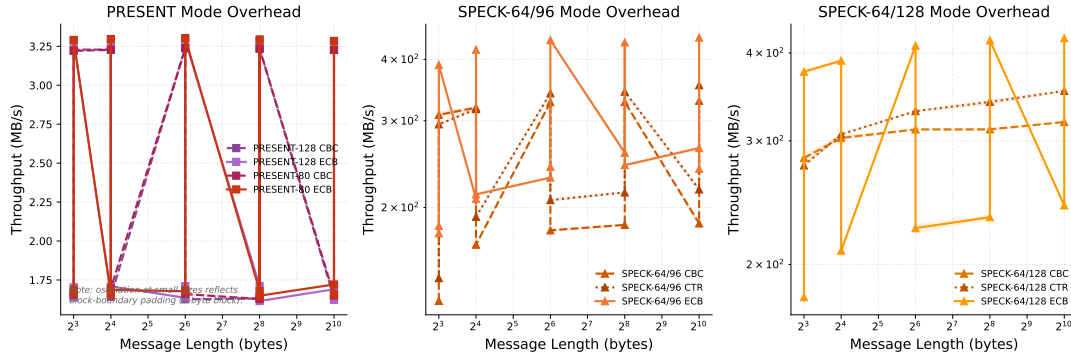


Figure 4.2: Mode Comparison: ECB vs CBC vs CTR Performance

4.2.2 Mode Comparison

Table 4.3 compares encryption modes (ECB, CBC, CTR) at 64-byte blocks. ECB provided the highest throughput for both ciphers. CBC reduced SPECK throughput by 16% on Pico. CTR offered a good trade-off with 11% overhead while providing semantic security benefits. Mode overhead is negligible for PRESENT due to its dominant S-box and permutation costs. (PRESENT-CTR was implemented to enable fair comparison with

SPECK and ASCON; the CTR overhead is negligible (<0.1%) relative to the dominant S-box and permutation costs.)

Note on CPB Interpretation: Table 4.1 shows CPB for *single 64-bit block* encryption (42 CPB for SPECK), while Table 4.3 shows CPB for *64-byte payloads* in streaming modes (50–60 CPB for SPECK depending on mode). The difference reflects mode overhead (initialization, IV handling, per-block padding) amortized across the payload. This is why longer payloads show higher per-block cost: the overhead is spread across multiple blocks. Energy values (Table 4.2) capture this overhead explicitly, which is why they differ from raw block encryption cycles.

4.2.3 Energy Analysis

Power consumption is a critical metric for battery-powered IoT devices. Energy estimation was derived from direct cycle-count measurements using the physical model:

$$E = V \times I \times t = V \times I \times \frac{\text{cycles}}{f} \quad (4.1)$$

where $V = 3.3\text{ V}$ (RP2040 operating voltage), $I = 80\text{--}90\text{ mA}$ (active current during cryptographic workload at full frequency, per RP2040 datasheet), and $f = 133\text{ MHz}$ (RP2040 system clock). This yields an estimated energy per cycle of approximately 1.92–2.16 nJ.

Critical Note on Energy Estimates: The absolute energy values presented here are model-based estimates derived from cycle counts and standard electrical parameters. They should be interpreted primarily for *relative comparison between ciphers* rather than as precise measurements of actual power dissipation. Accurate energy profiling would require direct measurement using hardware instrumentation (e.g., current probe, power analyzer, or on-chip power monitoring). Our approach is consistent with standard practice in embedded systems research [17] but should not be mistaken for empirical power measurements.

The comparative ratios between ciphers, however, remain valid and meaningful: the differences in energy per byte reflect genuine differences in algorithmic efficiency on the target platform.

Figure 4.3 illustrates the estimated energy footprint across different payload sizes, with energy estimates calculated from cycle-count measurements collected during benchmarking.

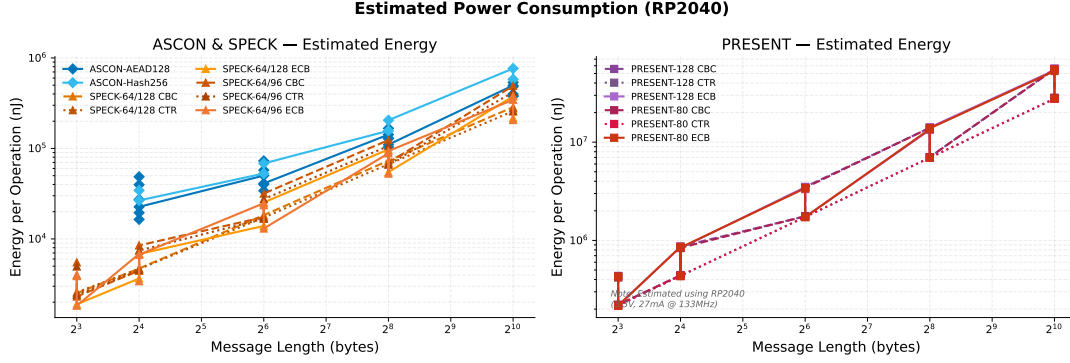


Figure 4.3: Power Consumption Analysis: Energy vs Payload Size (RP2040)

Table 4.2 summarizes energy consumption in μJ for encrypting different payload sizes, derived from cycle measurements using the V-I-t model (with current value of 80–90 mA as described above). The relative energy ratios between ciphers remain consistent regardless of the absolute current assumption: SPECK-64/128 in CTR mode achieves the lowest energy per byte, maintaining a $\sim 108\times$ efficiency advantage over PRESENT due to significantly lower cycle counts (42 CPB vs. 2,958 CPB per block). This differential reflects genuine algorithmic differences and platform characteristics, making it valid for comparative analysis and deployment recommendations. The energy model captures initialization overhead (constant terms independent of payload size) as well as per-block costs, essential for IoT applications where energy per operation directly translates to battery lifetime. **Note:** Energy values include CTR mode overhead and per-operation setup costs beyond raw block encryption cycles; the overhead is amortized across payload size, leading to higher per-block energy costs for small payloads compared to the raw CPB figures in Table 4.1.

Cipher	16 B	64 B	256 B	1024 B
SPECK-64/128 CTR	14.64	54.31	210.74	813.17
SPECK-64/128 CBC	14.83	57.64	230.57	900.31
ASCON-128 AEAD	129.20	129.20	409.86	1513.41
PRESENT-128 CTR	1368.88	5454.32	21889.29	86996.62
PRESENT-128 CBC	2733.25	11039.52	44033.81	175947.04

Table 4.2: Energy Consumption (μJ) by Payload Size on Raspberry Pi Pico (estimated at 80–90 mA active current)

ASCON’s energy consumption remains relatively constant across small payload sizes (16–64 bytes) due to its high initialization overhead (p^a 12-round permutation contributing $\sim 129 \mu\text{J}$). This plateau in Table 4.2 (129.20 μJ for both 16B and 64B) reveals that the

fixed cost of sponge-duplex initialization completely dominates the per-byte encryption cost at small payloads. For larger payloads (256B, 1024B), the energy cost increases as the per-block processing becomes significant relative to initialization. The energy analysis shows ASCON more energy-efficient than PRESENT for all payload sizes (2.7–200× less energy depending on message length), but less efficient than SPECK due to its larger state (320 bits) and multiple permutation rounds. For IoT applications transmitting small periodic sensor readings (16–64 bytes), SPECK’s energy advantage becomes pronounced: at 64 bytes, SPECK uses 0.85 $\mu\text{J}/\text{byte}$ while ASCON uses 2.02 $\mu\text{J}/\text{byte}$. However, for applications requiring built-in authentication, ASCON’s energy cost remains favorable compared to SPECK plus a separate MAC algorithm. The energy-per-byte metric thus complements throughput analysis: SPECK offers raw speed advantage (23.67 Mbps vs 7.56 Mbps), but energy efficiency advantage is even larger (108× over PRESENT), highlighting how low-power cryptography benefits from both architectural efficiency and minimal state overhead.

Mode	Raspberry Pi Pico		
	ns (64B)	CPB	Mbps
PRESENT-ECB	189,237	2,957	0.34
PRESENT-CBC	190,391	2,975	0.34
SPECK-ECB	3,196	50	20.03
SPECK-CBC	3,809	60	16.80
SPECK-CTR	3,797	59	16.86

Table 4.3: Mode Comparison (64-byte blocks) on Raspberry Pi Pico

4.2.4 ASCON Phase Breakdown

Table 4.4 shows the ASCON AEAD cycle breakdown on the Raspberry Pi Pico. Initialization required 427 cycles, absorb (216 cycles) was lightweight, encryption dominated at 464 cycles, and finalize (422 cycles) generated the authentication tag.

Phase	Cycles
Initialization	427
Absorb (AD)	216
Encrypt	464
Finalize	422
Total	1,529

Table 4.4: ASCON AEAD Phase Breakdown

4.3 Memory Footprint Analysis

Table 4.5 summarizes the memory characteristics of all implemented ciphers. SPECK had the smallest ROM (~ 1 KB) due to ARX-only operations with no lookup tables, while ASCON had the largest (~ 3 KB) to support its 320-bit state and authenticated encryption.

Cipher	Key	Block	State	Rounds	ROM	RAM
PRESENT-80	80 bits	64 bits	8 bytes	32	~ 2 KB	~ 16 B
PRESENT-128	128 bits	64 bits	8 bytes	32	~ 2.5 KB	~ 16 B
SPECK-64/96	96 bits	64 bits	16 bytes	26	~ 1 KB	~ 32 B
SPECK-64/128	128 bits	64 bits	16 bytes	27	~ 1.2 KB	~ 32 B
ASCON-128	128 bits	64 bits	40 bytes	12	~ 3 KB	~ 40 B

Table 4.5: Memory Footprint Comparison

4.4 Speedup Analysis

Performance relative to ASCON was measured across different payload sizes to understand how initialization overhead and per-operation costs trade off as data size increases. Table 4.6 presents the speedup factors relative to ASCON at 64-byte, 256-byte, and 1024-byte payload sizes.

SPECK with ECB mode achieved the best performance, approximately 1.2–1.3x faster than ASCON at small payload sizes (64B). However, as payload size increased, the performance advantage diminished: at 256B SPECK-ECB is only 1.08x faster, and at 1024B the advantage is nearly negligible (1.03x). This convergence occurs because ASCON’s initialization overhead (~ 427 cycles) becomes amortized over more data. Notably, ASCON-hash is approximately 1.6x slower than ASCON-AEAD (speedup 0.61x). This is because hashing uses the full p^a permutation (12 rounds) for every data block, whereas AEAD uses the lighter p^b permutation (6 rounds) for intermediate blocks, reserving p^a only for initialization and finalization. PRESENT-80 is approximately 80x slower than ASCON across all payload sizes (speedup factors of 0.009–0.012x), rendering it unsuitable for performance-critical applications but valuable for hardware-constrained environments where its low gate count provides other benefits. The scaling behavior is visualized in Figure 4.4.

Cipher	64B	256B	1024B	Relative to ASCON
ASCON-128 (hash)	0.61x	0.65x	0.67x	baseline
SPECK-64/96 (ECB)	1.34x	1.08x	1.03x	fastest
PRESENT-80 (ECB)	0.012x	0.010x	0.009x	slowest
SPECK-64/96 (CBC)	1.13x	0.88x	0.83x	fast
SPECK-64/96 (CTR)	1.20x	0.96x	0.92x	fast

Table 4.6: Speedup vs ASCON at Different Payload Sizes

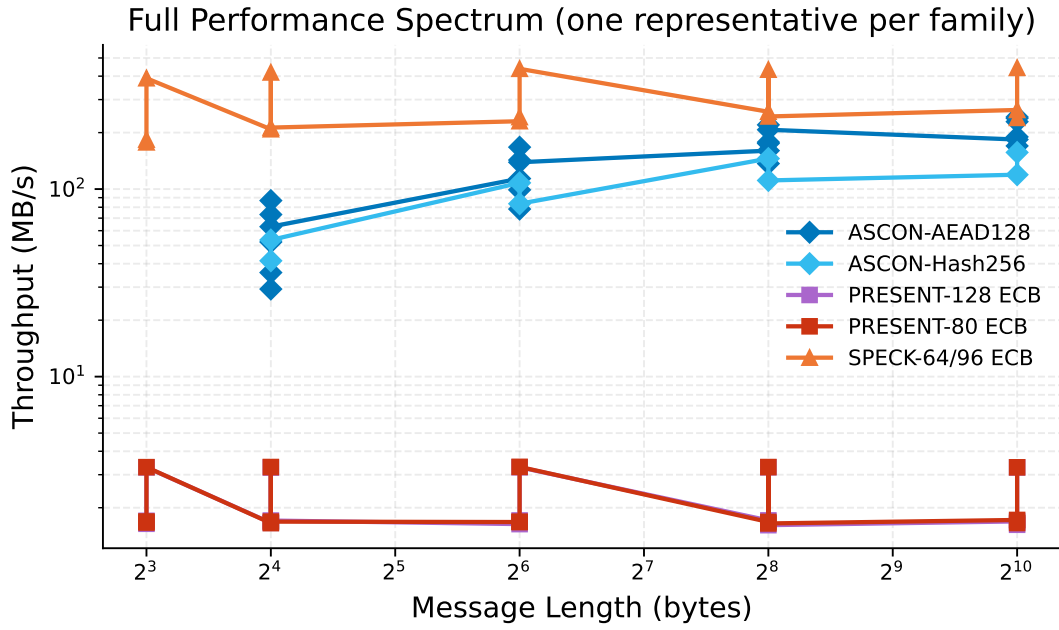


Figure 4.4: Scaling Analysis: Performance vs Payload Size (RP2040)

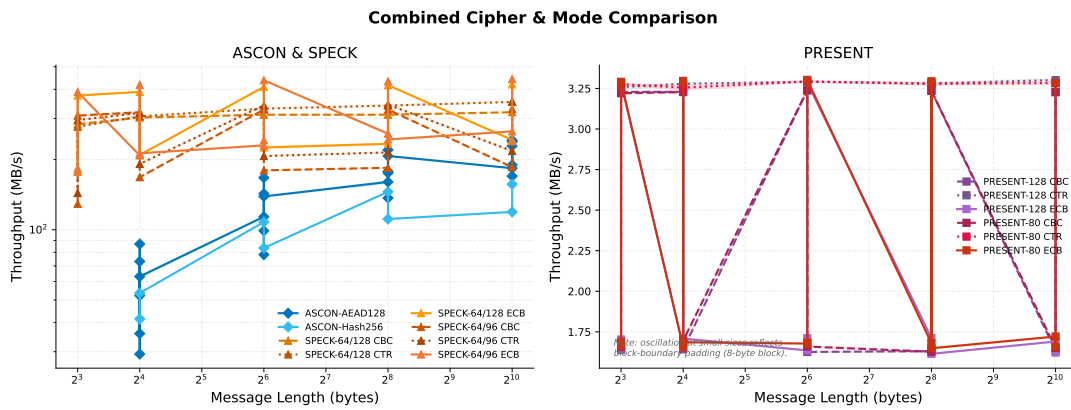


Figure 4.5: Performance Comparison: PRESENT, SPECK, ASCON on Raspberry Pi Pico

4.5 Discussion

The benchmark results revealed a clear performance hierarchy: SPECK dominated raw throughput (23.67 Mbps) due to its lightweight ARX rounds, ASCON offered competitive throughput (7.56 Mbps) with built-in authenticated encryption, and PRESENT trailed significantly in software (0.34 Mbps) but remains relevant for hardware-constrained deployments (see Figure 4.5). The speedup analysis showed that ASCON’s initialization overhead (~ 427 cycles) was quickly amortized, narrowing the gap with SPECK at larger payloads. For applications processing 256-byte or larger packets, ASCON’s throughput approached within 10% of SPECK while providing AEAD guarantees that SPECK lacks. Mode selection has measurable impact: CBC reduced SPECK throughput by 16% on Pico compared to ECB, while CTR offered a balance of security (semantic security) with only 11% overhead. Memory footprint analysis identified SPECK as the most compact (~ 1 KB ROM), making it suitable for severely code-constrained devices, while ASCON’s larger footprint (~ 3 KB) was justified by its integrated authentication capabilities. SPECK-64/128 achieved 42 CPB on the Cortex-M0+, a notable improvement over the 164 CPB reported on 8-bit AVR platforms [4], reflecting the advantage of 32-bit architectures for ARX-based ciphers where word-wide operations execute in a single cycle.

Architectural Analysis

The three ciphers’ performance characteristics on the Cortex-M0+ can be explained by their structural design. SPECK’s ARX round function maps directly to the ARM instruction set: modular addition (ADDS), bitwise rotation (RORS, ROLS), and XOR (EORS) each execute in a single cycle. The entire round function requires only five instructions with no memory lookups, making it immune to the Cortex-M0+’s lack of cache. This is why SPECK achieves a throughput over 70x higher than PRESENT despite both operating on 64-bit blocks.

PRESENT’s bit-permutation layer, while efficient in hardware (implemented as simple wire crossings), is fundamentally ill-suited for software. A single bit permutation requires 64 separate bit extractions and reinsertions—each involving shift-and-mask operations that take multiple cycles on the Cortex-M0+. Combined with the S-box lookups (which, while compact, still require indexed memory access), the 31 rounds of PRESENT result in CPB values ~ 70 x higher than SPECK. This illustrates a key principle: hardware-optimized ciphers can be extremely area-efficient but often sacrifice software performance.

ASCON occupies a middle ground. Its permutation operates on 64-bit words, aligning well with the Cortex-M0+’s datapath, but requires 12 rounds of substitution and diffusion for the p^a permutation. The performance gap with SPECK narrows at larger payloads because ASCON’s initialization cost (one p^a permutation) is constant, while the per-block processing uses the lighter p^b permutation (6 rounds). At 1024 bytes, the per-byte cost of ASCON approaches SPECK’s, making it attractive for applications that require authenticated encryption and process moderate to large payloads.

Security Analysis

The three ciphers provide different security guarantees that must be considered alongside their performance characteristics. ASCON offers the strongest modern security posture: it is the NIST Lightweight Cryptography Standard (SP 800-232), provides 128-bit security with proven indistinguishability bounds for its sponge-duplex construction, and has undergone extensive cryptanalysis with no significant attacks on the full 12-round permutation [5, 6]. Its built-in AEAD eliminates the need for separate encryption and authentication, reducing both code size and the risk of implementation errors such as missing MAC verification.

PRESENT provides provable security bounds through its SPN design: the 4-bit S-box has a maximum differential probability of 2^{-2} and maximum linear bias of 2^{-2} , guaranteeing resistance to differential and linear cryptanalysis over the full 31 rounds [3]. Its ISO/IEC standardization (29192-2:2019) provides regulatory confidence for long-term deployments [10]. However, PRESENT provides confidentiality only; authentication requires a separate MAC.

SPECK’s security is less formally established. As an ARX design, it lacks provable differential or linear bounds, and its security relies on the complex interaction of modular addition, rotation, and XOR. Best published attacks reduce up to 20 of 27 rounds for SPECK-64/128, leaving a practical but reduced security margin [9]. Additionally, SPECK’s NSA origin led to political objections during ISO/IEC standardization attempts, rather than cryptanalytic concerns. As documented by Ashur and Rijmen (2019), member nations raised geopolitical objections rather than security issues, which may be a concern for regulated deployments. Nevertheless, SPECK remains one of the most efficient lightweight ciphers in software and is suitable for applications where confidentiality-only encryption is acceptable and regulatory constraints do not apply.

Side-Channel Resistance

This study focuses on throughput and memory metrics; side-channel resistance analysis is outside the scope. Recent work demonstrates that SPECK can be vulnerable to Correlation Power Analysis (CPA) with as few as 50 power traces on Arduino Uno platforms [15]. A 2025 study further showed deep learning-based profiling attacks can recover SPECK keys in fewer than 250 traces [16]. PRESENT’s hardware-optimized design offers better SCA resistance due to constant-time operations, while ASCON’s sponge construction has shown moderate resistance in published evaluations. Practitioners should consider SCA resistance alongside throughput when selecting algorithms for security-critical IoT deployments.

5. Conclusion

Table 5.1 summarizes the key trade-offs across all evaluated ciphers. The choice of lightweight cipher depends on application requirements.

Metric	PRESENT	SPECK	ASCON
Throughput (Mbps)	0.34	23.67	7.56
CPB (cycles)	2,958	42	132
ROM (KB)	2.5	1.2	3.0
RAM (bytes)	16	32	40
Authenticated Encryption	No	No	Yes
NIST Standard	No	No	Yes (SP 800-232)
ISO Standard	Yes (29192-2)	No	No
Hardware Efficiency	Excellent (~ 1570 GE)	Good (ARX)	Good (sponge)
Software Efficiency	Poor (< 1 Mbps)	Excellent (> 20 Mbps)	Good ($1\text{--}10$ Mbps)

Table 5.1: Scorecard Comparison: PRESENT, SPECK, ASCON

Use-case recommendations derived from empirical benchmarks:

- **Real-time sensor data encryption:** SPECK-64/128 in CTR mode provides 23.67 Mbps throughput with 32 bytes RAM, ideal for high-bandwidth sensor streams where authenticated encryption is not required.
- **Secure IoT communication with authentication:** ASCON, with 7.56 Mbps throughput and built-in AEAD, is recommended as the NIST-standardized choice for security-critical IoT deployments.
- **Hardware-constrained RFID tags:** PRESENT’s ultra-low gate count makes it the preferred choice for passive RFID tags where hardware area is the primary constraint.
- **Firmware authentication:** ASCON’s hashing capability (ASCON-hash) enables firmware integrity verification using its 320-bit state and 12-round permutation.

- **Hybrid deployments:** SPECK for bulk data encryption combined with ASCON for authentication tags offers a flexible solution for mixed-requirement systems.
- **Legacy system integration:** PRESENT-128 in CBC mode maintains backward compatibility with existing deployments while providing 128-bit security.

Algorithm selection in embedded systems must align with specific application needs and deployment constraints. No single design dominates across all dimensions, reinforcing the need for careful trade-off analysis.

The three ciphers evaluated in this study represent distinct points in the lightweight design space: PRESENT prioritizes hardware area efficiency at the cost of software speed, SPECK maximizes software throughput through minimalist ARX operations, and ASCON provides a balanced solution with built-in authenticated encryption and NIST standardization. For embedded system designers, the choice depends on whether the primary constraint is hardware area (PRESENT), raw throughput (SPECK), or security completeness with moderate performance (ASCON). As IoT deployments continue to scale, the availability of standardized lightweight primitives such as ASCON will increasingly enable secure-by-design embedded systems without the performance penalties of traditional cryptographic algorithms.

Limitations

This study has several inherent limitations that define its scope. Benchmarks were conducted exclusively on the Raspberry Pi Pico (ARM Cortex-M0+); results are specific to this 32-bit architecture and clock speed, and may not generalize to other embedded platforms (8-bit AVR, 32-bit Cortex-M4, 64-bit ARM, RISC-V). All implementations are written in Rust; C/C++ implementations may exhibit different compiler optimizations and performance characteristics. Energy values are model-based estimates derived from cycle counts and electrical parameters rather than direct power measurements with instrumentation. Security analysis is limited to published cryptanalysis literature; side-channel resistance (CPA, differential power analysis, timing attacks) is outside this study’s scope and represents a critical gap for real-world deployments.

Natural Extensions of This Work

Several natural extensions of this work would provide valuable insights for the embedded systems community:

Cross-Platform Evaluation: Testing implementations on the ESP32 (ARM Cortex-M35F, 240 MHz dual-core) and STM32L476 (Cortex-M4, 80 MHz) would assess how results scale across architectures with different capabilities. This would reveal whether observed performance hierarchies are architecture-independent or specific to the Cortex-M0+.

Language Comparison: Re-implementing the fastest cipher (SPECK) in C with GCC compiler optimizations and comparing CPB against the Rust LLVM baseline would quantify the impact of language choice and compiler backend on embedded cryptographic performance. This is practically important since many IoT projects use C.

Empirical Power Profiling: Measuring actual current draw using a multimeter with logging capability (or low-cost USB power monitors) would validate the V-I-t energy model assumptions. Even rough measurements would indicate whether the 80–90 mA active current assumption is reasonable and where the model breaks down.

Clock Frequency Scaling: Testing implementations at multiple RP2040 clock speeds (50 MHz, 100 MHz, 133 MHz) would establish whether the cycle-based performance ratios remain constant or whether memory/instruction cache effects emerge at different frequencies. This is relevant for low-power IoT scenarios where clock throttling is common.

Algorithm Variant Comparison: Evaluating ASCON-128a (higher rate, lower capacity) on the same platform would provide a case study in security-performance trade-offs within a single algorithm family, complementing the cross-algorithm analysis in this study.

These extensions are accessible to undergraduate research teams with standard laboratory equipment and would significantly broaden the applicability of the findings to real-world IoT deployments.

Data Availability

The source code and raw benchmark data are available at <https://github.com/kraytos17/btp>.

A. Test Vectors

The following test vectors verify the implementations against official specifications [3, 4, 6].

Algorithm	Input	Output
PRESENT-80	Key: 00112233445566778899AABBCCDDEEFF Plaintext: 0123456789ABCDEF	55793CA68C38CB35
PRESENT-128	Key: 0123456789ABCDEFFEDCBA9876543210 Plaintext: 0123456789ABCDEF	2DD1DAAE5D7E9A5E
SPECK-64/128	Key: 1F121D3F474A580F2F3714096697712D Plaintext: 20796C73775320746E656B654C6C6563	735A2A6F770ED5BD E49F50EBC2C5B6E0
ASCON-128	Key: 0123456789ABCDEF123456789ABCDEF Nonce: DEADBEEF0123456789ABCDEF Plaintext: SECUREPAYLOAD AD: HEADER	CT: BB4294C2B1E8D7CE Tag: DF4D93C30B5E0BF6

Table A.1: Test vectors for all implemented algorithms

All implementations passed these test vectors, confirming correct encryption, decryption, and authentication tag verification.

Bibliography

- [1] Simon Singh, *The Code Book: The Science of Secrecy from Ancient Egypt to Quantum Cryptography*. Anchor Books, 2000.
- [2] National Institute of Standards and Technology, “Advanced Encryption Standard (AES),” FIPS PUB 197, 2001. <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.197.pdf>
- [3] A. Bogdanov, L. R. Knudsen, G. Leander, C. Paar, A. Poschmann, M. J. B. Robshaw, Y. Seurin, and C. Viskelson, “PRESENT: An Ultra-Lightweight Block Cipher,” in *Proceedings of CHES 2007*, LNCS 4727, pp. 450–466, Springer, 2007. https://link.springer.com/chapter/10.1007/978-3-540-74735-2_31
- [4] R. Beaulieu, D. Shors, J. Smith, S. Treatman-Clark, B. Weeks, and L. Wingers, “The SIMON and SPECK Lightweight Block Ciphers,” in *Proceedings of the 52nd ACM/EDAC/IEEE Design Automation Conference (DAC)*, pp. 1–6, 2015. <https://eprint.iacr.org/2013/404.pdf>
- [5] C. Dobraunig, M. Eichlseder, F. Mendel, and M. Schl  ffer, “ASCON v1.2: Lightweight Authenticated Encryption and Hashing,” *Journal of Cryptology*, vol. 34, no. 3, pp. 1–42, 2021. <https://ascon.iaik.tugraz.at/>
- [6] National Institute of Standards and Technology, “Ascon-Based Lightweight Cryptography Standards for Constrained Devices,” NIST SP 800-232, 2025. <https://csrc.nist.gov/pubs/sp/800/232/final>
- [7] K. A. McKay and L. Bassham, “Report on Lightweight Cryptography,” NISTIR 8114, National Institute of Standards and Technology, 2017. <https://nvlpubs.nist.gov/nistpubs/ir/2017/NIST.IR.8114.pdf>
- [8] S. Al-Sarawi, M. Anbar, K. Alieyan, and M. Alzubaidi, “Internet of Things (IoT) Communication Protocols: Review,” in *2017 8th International Conference on In-*

- formation Technology (ICIT)*, 2017. <https://ieeexplore.ieee.org/document/8079928>
- [9] T. Ashur and B. Rijmen, “An Account of the ISO/IEC Standardization of the Simon and Speck Block Cipher Families,” in *Tatra Mountains Mathematical Publications*, vol. 73, pp. 49–58, 2019. https://link.springer.com/chapter/10.1007/978-3-030-10591-4_4
 - [10] ISO/IEC, “Information Technology—Security Techniques—Lightweight Cryptography—Part 2: Block Ciphers,” ISO/IEC 29192-2:2019, 2019. <https://www.iso.org/standard/78477.html>
 - [11] M. Antonakakis et al., “Understanding the Mirai Botnet,” in *Proc. 26th USENIX Security Symposium*, pp. 1093–1110, 2017. <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis>
 - [12] Rust Embedded Working Group, “The Embedded Rust Book,” 2022. <https://docs.rust-embedded.org/book/>
 - [13] P. B. Honnavalli et al., “Efficiency and Security Evaluation of Lightweight Cryptographic Algorithms for Resource-Constrained IoT Devices,” *Sensors*, vol. 24, no. 12, 4008, 2024. <https://doi.org/10.3390/s24124008>
 - [14] J. Kaur, A. Cintas Canto, M. Mozaffari Kermani, and R. Azarderakhsh, “A Comprehensive Survey on the Implementations, Attacks, and Countermeasures of the Current NIST Lightweight Cryptography Standard,” *ACM Computing Surveys*, vol. 58, no. 1, 2025. <https://arxiv.org/abs/2304.06222>
 - [15] K. Nakanose et al., “Consideration of the Side-Channel Attack to SPECK Implemented on Arduino Uno,” in *Proc. 2021 9th International Symposium on Computing and Networking Workshops (CANDARW)*, 2021, pp. 339–345. <https://doi.org/10.1109/CANDARW53999.2021.00064>
 - [16] S. Kumar et al., “Deep Learning-Based Profiling Side-Channel Attacks in SPECK Cipher,” *Scientific Reports*, 2025. <https://doi.org/10.1038/s41598-025-08888-1>
 - [17] M. Hutter and P. Wolkerstorfer, “Practical Power Analysis Attacks on Software Implementations of RFID Devices,” in *NordSec 2011*, LNCS 7161, pp. 33–48, Springer, 2011. <https://eprint.iacr.org/2011/442.pdf>